

## Subject Description Form

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| <b>Subject Code</b>                                   | EE2004                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Subject Title</b>                                  | Electrical Energy Systems Fundamentals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Credit Value</b>                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Level</b>                                          | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Pre-requisite/<br/>Co-requisite/<br/>Exclusion</b> | Pre-requisite for EE2004: EE2002<br>Pre-requisite for EE2004A: EE2002A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Objectives</b>                                     | <ol style="list-style-type: none"> <li>1. To provide an overview of the supply, utilization, and control of electrical energy.</li> <li>2. To introduce energy issues, and assist students in placing these topics and technologies in perspective.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Intended Learning Outcomes</b>                     | <p>Upon completion of the subject, students will be able:</p> <ol style="list-style-type: none"> <li>a. To master the fundamental knowledge on electrical energy systems.</li> <li>b. To identify, analyze, and solve technical problems using mathematics and engineering techniques.</li> <li>c. To be aware of equipment characteristics in modern electrical power systems.</li> <li>d. To be able to conduct laboratory work in teams and present the findings.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b>      | <ol style="list-style-type: none"> <li>1. <b><i>Nature of electrical energy system:</i></b> Power system definition, layout and basic components, transmission and distribution structure, role of transformers. The interconnected power system. HVDC transmission. Layout of a substation, distribution structure, overhead lines and cables, circuit breaking, protection concepts, line protection.</li> <li>2. <b><i>Generation &amp; energy:</i></b> Principles of energy conversion, power plant and busbar layout, types of generators and turbines. Concept of generation control and operating chart. Pumped storage and wind turbine. Renewable and non-renewable sources. Sustainable development.</li> <li>3. <b><i>Tariffs:</i></b> Concept and structure of electricity market. Concepts of tariff design. Tariff structures. Conventional and new tariffs in different utilities. Two-part tariff, introduction to deregulation and load management concepts.</li> <li>4. <b><i>Power system basic principles:</i></b> Concept of phasor, representation and properties of phasor. Inductive and capacitive circuit. Real and reactive power. Single and three phase systems. Per-phase analysis. Per unit system and calculation. Power factor correction.</li> <li>5. <b><i>Transformers:</i></b> Construction and operating principles. Equivalent circuits. Tests on transformers. Voltage regulation and power efficiency. Parallel operation. Three-phase transformers and phase grouping. Autotransformers. Instrument transformers.</li> <li>6. <b><i>Lines &amp; cables:</i></b> Overhead line construction including transposition and bundling. Primary (RLCG) and general (ABCD) parameter calculations. Line equations and performance calculations. Corona loss and interference. Cable types, construction, electrical stress, and thermal characteristics.</li> </ol> <p><b>Industry Lectures:</b><br/>Introduction to power utilities and their operational model<br/>Introduction to generation, transmission and distribution</p> <p><b>Laboratory Experiment:</b><br/>Experiments on single phase transformer.<br/>Experiments on three-phase transformer.<br/>Computer exercises on transmission line parameters calculations.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                  | <b>Case study:</b><br>Intermittent energy resources and major issues with their integration into power grids<br>Application of voltage source converter technology in power systems<br>Smart grids and the coordination of behind-the-meter technologies (EV, PV, storage)<br>Autonomous energy grids and their applicability in Hong Kong<br>Offshore wind power generation, overall global potential vs. global energy demand<br>Battery energy storage systems and their applications in power systems |             |                                                   |   |   |   |
| <b>Teaching/Learning Methodology</b>                                                                                                                                                                                                                                                                                                                             | Lectures are the primary means of conveying the basic concepts and knowledge, teaching students the skills in identifying, analyzing, and solving technical problems and providing students feedback in relation to their learning. Laboratory experiments and case studies are designed, as supplement to the lecturing materials, for students to gain practical experiences and be aware of equipment characteristics and environment issues on the modern electrical power system.                    |             |                                                   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                  | Teaching/Learning Methodology                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Outcomes    |                                                   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | a           | b                                                 | c | d |   |
|                                                                                                                                                                                                                                                                                                                                                                  | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ✓           | ✓                                                 | ✓ |   |   |
|                                                                                                                                                                                                                                                                                                                                                                  | Case studies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ✓           | ✓                                                 | ✓ |   |   |
| Experiments                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             | ✓                                                 | ✓ |   |   |
| <b>Assessment Methods in Alignment with Intended Learning Outcomes</b>                                                                                                                                                                                                                                                                                           | Specific assessment methods/tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | % weighting | Intended subject learning outcomes to be assessed |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             | a                                                 | b | c | d |
|                                                                                                                                                                                                                                                                                                                                                                  | 1. Examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 60%         | ✓                                                 | ✓ | ✓ |   |
|                                                                                                                                                                                                                                                                                                                                                                  | 2. Class tests                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 18%         | ✓                                                 | ✓ | ✓ |   |
|                                                                                                                                                                                                                                                                                                                                                                  | 3. Lab performance and report                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 10%         |                                                   |   | ✓ | ✓ |
|                                                                                                                                                                                                                                                                                                                                                                  | 4. Case studies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 12%         | ✓                                                 | ✓ | ✓ |   |
|                                                                                                                                                                                                                                                                                                                                                                  | Total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 100%        |                                                   |   |   |   |
| The outcomes on concepts, design and applications are assessed by examinations and tests whilst those on analytical skills, problem solving techniques and practical considerations of electrical energy systems, as well as team work and technical report writing abilities are evaluated by lab performance and reports, and assignment / case study reports. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                                                   |   |   |   |
| <b>Student Study Effort Expected</b>                                                                                                                                                                                                                                                                                                                             | Class contact:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                  | ▪ Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             | 33 Hrs.                                           |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                  | ▪ Laboratory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             | 6 Hrs.                                            |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                  | Other student study effort:                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                                   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                  | ▪ Laboratory preparation / Report                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             | 9 Hrs.                                            |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                  | ▪ Case study / Self-study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             | 57 Hrs.                                           |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                  | Total student study effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             | 105 Hrs.                                          |   |   |   |
| <b>Reading List and References</b>                                                                                                                                                                                                                                                                                                                               | <b>Textbooks:</b><br>1. J. Grainger, W. D. Stevenson, Power System Analysis, McGraw-Hill, Latest edition                                                                                                                                                                                                                                                                                                                                                                                                  |             |                                                   |   |   |   |

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|  | <ol style="list-style-type: none"><li>2. B. M. Weedy, B. J. Cory, N. Jenkins, J. B. Ekanayake, G. Strbac, Electric Power Systems, Wiley, 5<sup>th</sup> Edition, Wiley, 2012</li><li>3. M. E. El-Hawary, Electrical Energy Systems, 2<sup>nd</sup> Edition, CRC Press, 2018</li></ol> <p><b>Reference books:</b></p> <ol style="list-style-type: none"><li>1. H. Saadat, Power System Analysis, 3<sup>rd</sup> Edition, PSA Publishing LLC, 2011</li><li>2. A. R. Bergen, V. Vittal, Power System Analysis, 2<sup>nd</sup> Edition, Pearson, 2000</li><li>3. J.D. Glover, M. S. Sarma, T.J. Overbye, Power System Analysis and Design, 6<sup>th</sup> Edition, Cengage Learning, 2016</li><li>4. D.P. Kothari, I.J. Nagrath, Modern Power System Analysis, McGraw-Hill, 4<sup>th</sup> Edition, 2011</li></ol> |
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